

Portfolio Expected Return and Variance

*Two-Asset Portfolio

$$R_{pt} = w_1 R_{1t} + w_2 R_{2t}$$

where R_{pt} is the return of portfolio at time t

w_i is the weight of investment in asset (company) i

R_{it} is the return of asset (company) i at time t

$$\begin{aligned}\text{Expected Return} &= E(R_{pt}) = \mu_p = E[w_1 R_{1t} + w_2 R_{2t}] \\ &= w_1 E(R_{1t}) + w_2 E(R_{2t}) \\ &= w_1 \mu_1 + w_2 \mu_2\end{aligned}$$

where μ_i is the mean of asset (company) i

$$\begin{aligned}\text{Variance} &= \text{Var}(R_{pt}) = \sigma_p^2 = \text{Var}(w_1 R_{1t} + w_2 R_{2t}) \quad \text{Note that } \text{Var}(X) = E[(X - \mu)^2] \\ &= E[(w_1 R_{1t} + w_2 R_{2t}) - (w_1 \mu_1 + w_2 \mu_2)]^2 \\ &= E[w_1 (R_{1t} - \mu_1) + w_2 (R_{2t} - \mu_2)]^2 \\ &= E[w_1^2 (R_{1t} - \mu_1)^2 + w_2^2 (R_{2t} - \mu_2)^2 + 2w_1 w_2 (R_{1t} - \mu_1)(R_{2t} - \mu_2)] \\ &= w_1^2 E(R_{1t} - \mu_1)^2 + w_2^2 E(R_{2t} - \mu_2)^2 + 2w_1 w_2 E[(R_{1t} - \mu_1)(R_{2t} - \mu_2)] \\ \sigma_p^2 &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_{12}\end{aligned}$$

where σ_i^2 is the variance of asset (company) i

σ_{ij} is the covariance between asset i and j

Or in matrix form

$$\text{Expected Return} = E(R_{pt}) = \mu_p = w_1 \mu_1 + w_2 \mu_2 = W^T \mu$$

$$\text{Variance } \text{Var}(R_{pt}) = \sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_{12} = W^T \Sigma W$$

$$\text{where } \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}; W = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}; W^T = [w_1 \quad w_2]; \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$$

*Three-Asset Portfolio

$$R_{pt} = w_1 R_{1t} + w_2 R_{2t} + w_3 R_{3t}$$

$$\begin{aligned}\text{Expected Return} &= E(R_{pt}) = \mu_p = E[w_1 R_{1t} + w_2 R_{2t} + w_3 R_{3t}] \\ &= w_1 \mu_1 + w_2 \mu_2 + w_3 \mu_3\end{aligned}$$

$$\begin{aligned}\text{Variance} &= \text{Var}(R_{pt}) = \sigma_p^2 = \text{Var}(w_1 R_{1t} + w_2 R_{2t} + w_3 R_{3t}) \\ &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + w_3^2 \sigma_3^2 + 2w_1 w_2 \sigma_{12} + 2w_2 w_3 \sigma_{23} + 2w_1 w_3 \sigma_{13}\end{aligned}$$

***N-asset Portfolio**

$$R_{pt} = \sum_{i=1}^N w_i R_{it}$$

$$\begin{aligned}\text{Expected Return} &= E(R_{pt}) = \mu_p = E\left[\sum_{i=1}^N w_i R_{it}\right] \\ &= \sum_{i=1}^N E[w_i R_{it}] = \sum_{i=1}^N w_i E[R_{it}] \\ &= \sum_{i=1}^N w_i \mu_i\end{aligned}$$

$$\begin{aligned}\text{Variance} &= \text{Var}(R_{pt}) = \sigma_p^2 = \text{Var}\left(\sum_{i=1}^N w_i R_{it}\right) \\ &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + \dots + w_N^2 \sigma_N^2 + 2w_1 w_2 \sigma_{12} + 2w_1 w_3 \sigma_{13} + \dots + 2w_1 w_N \sigma_{1N} + \dots + 2w_{N-1} w_N \sigma_{N-1,N} \\ &= \sum_{i=1}^N w_i^2 \sigma_i^2 + \sum_{i=1}^N \sum_{\substack{j=1 \\ i \neq j}}^N w_i w_j \sigma_{ij} \quad \text{Note that } \sigma_{ii} = \sigma_i^2 \\ &= \sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{ij}\end{aligned}$$

Or in matrix form

$$\text{Expected Return} = E(R_{pt}) = \mu_p = \sum_{i=1}^N w_i \mu_i = W^T \mu$$

$$\text{Variance } \text{Var}(R_{pt}) = \sigma_p^2 = \sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_{ij} = W^T \Sigma W$$

$$\text{where } \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_N \end{bmatrix}; W = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_N \end{bmatrix}; W^T = [w_1 \quad w_2 \quad \dots \quad w_N]; \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1N} \\ \sigma_{21} & \sigma_{22} & & \sigma_{2N} \\ \vdots & & \ddots & \vdots \\ \sigma_{N1} & \sigma_{N2} & \dots & \sigma_{NN} \end{bmatrix}$$